

CHEMISTRY (THEORY) – MOCK PAPER SET 3

(CHALLENGE LEVEL: FOR TOP SCORERS)

Time Allowed: 3 Hours

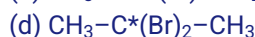
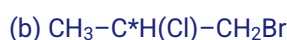
Maximum Marks: 70

By Pragya • Analyzed from Sets 56/5/1, 56/7/1, and HFG1E/5

SECTION A (Tricky MCQs – 1 Mark Each)

1.

Which of the following molecules has a chiral center correctly labelled?



Source: 56/5/1 HFG – Identifying chirality in chains

2.

1 mole of liquid A and 2 moles of liquid B make a solution having a total vapour pressure of 40 torr. Pure A and B pressures are 45 and 30 torr. The solution:

(a) Is an ideal solution

(b) Shows positive deviation

(c) Shows negative deviation

(d) Is a maximum boiling azeotrope

Source: 56/5/1 HFG – Calculation required

3.

The magnetic moment of $[\text{NiCl}_4]^{2-}$ is approximately:

(a) 1.82 BM

(b) 2.82 BM

(c) 4.42 BM

(d) 5.46 BM

Source: 56/5/1 HFG – Weak ligand field consideration

4.

For $\text{A} + \text{B} \rightarrow \text{C} + \text{D}$, doubling A leaves rate constant. Doubling B makes rate 4 times. The orders are:

(a) A = 1, B = 2

(b) A = 0, B = 2

(c) A = 0, B = 1

(d) A = 2, B = 0

Source: 56/5/1 HFG Rate Data Table

5.

Which of the following alcohols will NOT undergo oxidation?

(a) Butanol

(b) Butan-2-ol

(c) 2-Methylbutan-2-ol

(d) 3-Methylbutan-2-ol

6.

Assertion (A): Boiling point of o-dichlorobenzene > p-isomer but melting point of para-isomer > ortho-isomer.

Reason (R): Para isomer has a symmetrical crystal structure fitting better in the lattice.

(a) A and R are true, R is correct explanation

Source: 56/5/1 HFG

SECTION B (Reasoning – 2 Marks Each)

7.

Chemical Kinetics: Double rate on 10K increase from 298K. Calculate Activation Energy (E_a).

Source: Variation of 56/5/1 RP

8.

Solutions: What deviation type is shown by Ethanol + Acetone? Give reason in terms of IMF.

9.

Inorganic Reasoning: (i) Why is separation of Lanthanoids difficult? (ii) Why is Eu^{2+} a strong reducing agent?

10.

Organic Mechanism: Why does racemisation occur in $\text{S}_{\text{N}}1$ reactions? Draw intermediate.

11.

Coordination: Draw structure of optical isomer of $[\text{Cr}(\text{C}_2\text{O}_4)_3]^{3-}$. Note magnetic nature.

SECTION C (Complex Problems — 3 Marks Each)

12.

Kinetics: azoisopropane decomposition. Data: $t=0$, $P=35.0$; $t=360$, $P=54.0$. Calculate k .

Source: Similar to 56/7/1 numerical

13.

Solutions: 0.3g Acetic acid in 30g benzene, $\Delta T_f = 0.45^\circ\text{C}$. Calculate % association (dimer). $K_f = 5.12$.

14.

Electrochemistry: Resistance 0.1M KCl = 100 Ω . 0.02M KCl = 520 Ω . Calculate conductivity and Λ_m (κ of 0.1M = 1.29 S/m).

15.

Organic Puzzle: A ($\text{C}_4\text{H}_9\text{I}$) + alc. KOH $\rightarrow$ B; B + HI $\rightarrow$ C (isomer of A). A + Na/ether $\rightarrow$ D (C_8H_{18}). Identify A-D.

16.

Mechanism: Write mechanism of acid-catalysed hydration of ethene.

17.

CFT: (i) Electronic configuration for d^4 Strong Field. (ii) Why $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ loses color on heating?

18.

d-Block Trends: (i) Basic vs acidic oxides. (ii) Cobalt(II) stability in aqueous vs complex environment.

SECTION D (Case-Based — 4 Marks Each)

19.

Case Study: Solvent Effects on Substitution

"In $\text{S}_\text{N}1$, polarity increases from reactant to transition state... polar solvent accelerates. In $\text{S}_\text{N}2$, polar solvent stabilizes nucleophile, increasing E_a ..."

(a) Why water ($\epsilon=79$) shows 300,000x faster rate than ethanol ($\epsilon=24$) for t-butyl chloride? (b) Arrange $\text{S}_\text{N}2$ reactivity: CH_3I , CH_3F , CH_3Br , CH_3Cl . (c) Why ethanol < water polarity?

20.

Case Study: Fuel Cells

Hydrogen-Oxygen fuel cells are highly efficient for space missions.

(a) Overall $\text{H}_2\text{-O}_2$ reaction. (b) Calculate ΔG if potential = 1.23 V. (c) Efficiency vs thermal power plants?

SECTION E (Long Answer – 5 Marks Each)

21.

(a) Electrochemistry:

(i) Calculate emf of $\text{Zn} \mid \text{Zn}^{2+}(0.1\text{M}) \parallel \text{H}^+(0.01\text{M}) \mid \text{H}_2(1 \text{ bar})$. (ii) Electricity (Faraday) to reduce 1 mole $\text{Cr}_2\text{O}_7^{2-}$ to Cr^{3+} ?

22.

(a) Organic Synthesis:

(i) A ($\text{C}_9\text{H}_{10}\text{O}$) + 2,4-DNP (+), Tollens (+), Cannizzaro (+). Vigorous oxidation $\rightarrow$ 1,2-benzenedicarboxylic acid. Identify A. (ii) pK_a : $\text{F-CH}_2\text{-COOH}$ vs $\text{Cl-CH}_2\text{-COOH}$? (iii) Stephen reaction on CH_3CN .

23.

(a) d- & f-Block:

(i) Graphs: Tungsten (W) vs Chromium (Cr) melting points & Manganese (Mn) dip. (ii) $\text{MnO}_4^- + \text{S}_2\text{O}_3^{2-} + \text{H}_2\text{O} \rightarrow ?$ (iii) Actinoid vs Lanthanoid contraction?

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High-Probability Questions for CBSE 2026